CHAPTER 68

Bariatric and metabolic surgery

Learning objectives

To know and understand:

- How to treat obesity as a disease
- Rationale for surgery and the concept of metabolic surgery
- Eligibility and NICE guidelines
- Multidisciplinary assessment and multimodal treatment
- The common operations and how they work
- How to assess and treat perioperative complications
- Follow-up, nutritional supplements and biochemical monitoring

INTRODUCTION

Obesity is becoming the plague of the twenty-first century. With overweight becoming the norm in most Western countries and developing countries, two-thirds of adults suffer from overweight or obesity (***Table 68.1***). Every clinician and definitely every surgeon faces the condition and its associated diseases, such as type 2 diabetes, as part of their practice. According to the World Health Organization (WHO), overweight and obesity are defined as abnormal or excessive fat accumulation that may impair health. For adults, WHO defines overweight as a body mass index (BMI) of 25 kg/m^2 or more and obesity as a BMI of 30 kg/m^2 or more. Severe obesity increases the risk of cancer, is associated with multiple other diseases, affects quality of life and reduces life expectancy by 5–20 years. Severe and complex obesity is a phrase commonly used for patients with BMI ≥35 kg/m^2 and obesity-related disease, or BMI ≥40 kg/m^2 by itself (***Table 68.2***).

Overweight and obesity can be considered normal physiological responses to the current food environment. Few people with obesity have a single identifiable genetic or hormonal basis. *MC4R* deficiency represents the most common genetic form of severe obesity, with heterozygous mutations in *MC4R* detected in up to 5% of patients with severe, early-onset obesity. Surgeons encounter the challenge of obesity on a daily basis as it affects the treatment of nearly every abdominal pathology in terms of approach and outcomes.

Obesity is a heterogeneous disease and the response of individuals seeking treatment to different therapeutic modalities is variable. Currently there are no available robust tools to predict this response. Therefore a trial of options is required. The principles of therapeutic interventions for all other diseases are applicable, including escalation of treatment, cessation of modalities that are not effective and addition of therapy when the response is insufficient or transient. Lifestyle modifications, supervised interventions, pharmacotherapy, bariatric surgery and bariatric surgery combined with pharmacotherapy are available interventions. It is important to stress that the response is a biological phenomenon and not a volitional one.

TABLE 68.1 Definitions of overweight and obesity.[a,b]

Adult weight status	BMI (kg/m^2)
Normal	18.5–24.9
Overweight	25.0–29.9
Class 1 obesity	30.0–34.9
Class 2 obesity	35.0–39.9
Class 3 obesity	≥40.0

Body mass index (BMI) = weight (kg)/height (m)^2
[a]Obesity for children is defined as BMI at or above the 95th centile.
[b]'Super-obesity' is a term commonly used to describe BMI ≥49.9 kg/m^2.

TABLE 68.2 Conditions that are associated with severe and complex obesity.

Type 2 diabetes
Hypertension
Dyslipidaemia
Obstructive sleep apnoea
Arthritis and functional impairment
Gastro-oesophageal reflux disease
Non-alcoholic fatty liver disease/non-alcoholic steatohepatosis
Polycystic ovary syndrome
Clinical depression
Various cancers, in particular endometrial cancer

Bariatric surgery comes from the Greek 'baros' (meaning weight/pressure) and 'iatric' (the medicine or surgery thereof).

Bariatric surgery is the branch of surgery involving manipulation of the stomach and/or small bowel to achieve weight loss and control of obesity-related disease.

RATIONALE

Bariatric surgery leads to weight loss of 25–35% of body weight (usually at least 15 kg) after 1 year, and sustained weight loss maintenance at 15–25% after 20 years. Additional benefits are that most or all of the obesity-related diseases improve as weight is lost and even independently of weight loss. Quality of life improves. A number of randomised controlled trials (RCTs) have reported on the outcomes of bariatric surgery versus intensive lifestyle interventions, and all favour surgery. The longest follow-up in an RCT is 5 years. However, the non-randomised Swedish Obese Subjects (SOS) study has now shown sustained weight loss and improvement in obesity-related disease up to 20 years after surgery. In this study 2010 patients who chose to have surgery were compared with 2037 controls who did not.

When the SOS study was conceived in 1987 all surgery was done by laparotomy and it was considered unethical to attempt to randomise patients between best medical therapy and bariatric surgery. All had best medical care throughout and follow-up was better than 99%. The SOS study was among the first to demonstrate that bariatric surgery also leads to survival benefit. The primary end point was overall mortality and a significant difference was found at a mean follow-up of 10 years. Many other studies have now found similarly, including one of nearly 8000 operated patients from Utah.

The Utah study was the first large study to show improved survival after gastric bypass surgery compared with matched population controls. The SOS study also reported a lower incidence of both microvascular and macrovascular complications at 15 years of follow-up in the surgical group. The Swedish registry (SOREG) data indicate lower mortality within only 3–4 years after surgery in patients with type 2 diabetes. Both the SOS and Utah studies have shown that bariatric surgery also effectively reduces cancer risk in large patient cohorts.

Rationale for surgery

Owing to the tendency for basal metabolic rate to decrease with dietary calorie restriction most people will regain all their weight, returning to the previous homeostatic set point. Bariatric surgery appears to alter this mechanism and 'reset' this point, with 15–25% weight loss maintenance for up to 20 years. Despite the fact that bariatric surgery leads to long-term survival benefit and improves obesity-related disease and quality of life, it is available to only a fraction of the individuals who could potentially benefit. This is largely because of inequalities in healthcare prioritisation, misconceptions and obesity stigma. Person-first language should always be used ('patient with obesity' or 'patient with diabetes') to avoid categorising patients as the disease, to reduce stigma.

Summary box 68.1

Rationale for surgery

- Because of the tendency for basal metabolic rate to decrease with dieting, most people will regain all their weight, returning to the previous homeostatic set point
- Bariatric surgery appears to alter this mechanism and 'reset' this point, with 15–25% weight loss maintenance for up to 20 years
- Bariatric surgery leads to long-term survival benefit and improves obesity-related disease and quality of life

METABOLIC SURGERY

The phrases 'metabolic' or 'diabetes' surgery are increasingly being used in conjunction with, or instead of, 'bariatric surgery' owing to the highly effective way that surgery improves the metabolic syndrome, with weight loss being a welcome additional effect. Type 2 diabetes is part of the 'metabolic syndrome', which includes high blood pressure, dyslipidaemia and polycystic ovary syndrome.

Control of type 2 diabetes improves with weight loss owing to an improvement in insulin resistance. Remarkably, diabetes control appears to improve after several types of bariatric surgery *before* meaningful weight loss occurs. Some of the effects on glucose metabolism can be attributed to caloric restriction, but changes in gut hormones levels, particularly glucagon-like-peptide 1 (GLP-1), have provoked much interest. GLP-1 is an incretin, a gut hormone that stimulates the beta cells in the pancreas to restore the normal first-phase insulin response after eating. Bile acids are also involved in this.

Although type 2 diabetes is a chronic disease and bariatric surgeons in the early 2000s initially claimed to 'cure' it, the emphasis has now changed to improving glycaemic control by lowering glycated haemoglobin (HbA1c) and improving insulin resistance, with the ultimate goal of reducing cardiovascular risk and improving survival. The term diabetes 'remission' is now commonly used by bariatric surgeons and endocrinologists, and is defined as patients being off all medication with normal glucose homeostasis. In the SOS study, patients with diabetes went into remission after surgery, and in patients without diabetes there was a decreased incidence of patients developing diabetes.

Summary box 68.2

Metabolic surgery

- The term **metabolic surgery** refers to the marked effects of surgery on diabetes and the metabolic syndrome, which may have a more important impact than weight loss itself
- Improvement in type 2 diabetes may be additional to weight loss
- Surgery is very cost-effective

COST-EFFECTIVENESS

A 2009 Health Technology Assessment report in the UK showed bariatric surgery to be cost-effective compared with non-surgical options. The incremental cost-effectiveness ratio (ICER) compared with no surgery was between £2000 and

£4000 per quality-adjusted life year (QALY) gained for patients with BMI ≥40 kg/m^2 over 20 years. For patients with BMI between 30 and <40 kg/m^2 the ICER was £1367 per QALY gained. Regarding maximum willingness to pay, compared with non-surgical interventions, if a decision-maker is willing to pay £20 000 for an additional QALY, then the probability of surgery being cost-effective over a 20-year time horizon was reported as 100%. The ICERs are similar to the cost-effectiveness of stopping smoking and routine statin therapy for the primary prevention of cardiovascular disease. In practice it means that the cost of the operation is recouped within 1–2 years after surgery from reduced medication costs. All the cost-effectiveness studies assess direct or indirect healthcare costs but not the additional benefits of surgery. Thus, return to paid work, coming off state benefits, improved functional capacity and quality of life are 'add-ons' that incur no cost. A systematic review including studies up to 2018 confirmed that bariatric surgery is cost saving over a lifetime. Medication costs for obesity-related comorbidities are substantially reduced after bariatric surgery in the shorter term.

ELIGIBILITY

Eligibility criteria were first proposed by the US National Institutes of Health in 1991, when the obesity epidemic was first recognised. All bariatric surgery was done by open laparotomy, and the safety profile was very different. The National Institute for Health and Care Excellence (NICE) in the UK recommends consideration of bariatric surgery for people with severe obesity in whom all non-surgical measures have been tried with no adequate weight loss achieved or maintained. Commitment to long-term follow-up and behaviour change are also advised. After a review of recent RCTs, NICE updated its guidance and lowered the BMI threshold to 30 kg/m^2 for recent-onset type 2 diabetes (***Table 68.3***). The criteria according to the International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO) Asia Pacific chapter for Asian patients include lower BMI thresholds because of the susceptibility of Asian populations to type 2 diabetes at a lower BMI.

TABLE 68.3 Summary of updates to National Institute for Health and Care Excellence (NICE) guidance on bariatric surgery, 2014 (CG189).

Bariatric surgery is a treatment option for anyone with BMI ≥40 kg/m^2
Offer an expedited assessment for people with BMI ≥35 kg/m^2 with onset of type 2 diabetes in past 10 years
Consider an assessment for people with BMI of 30–34.9 kg/m^2 with onset of type 2 diabetes within 10 years
Consider an assessment for people of Asian origin with onset of type 2 diabetes at a lower BMI than other populations
Bariatric surgery is the option of choice for adults with BMI >50 kg/m^2 when other interventions have not been effective
People fitting the above criteria are also required to be receiving or to receive assessment in a specialist weight management service before referral to a surgical team

BMI, body mass index.

TABLE 68.4 National Institute for Health and Care Excellence (NICE)-accredited guidance on the make-up of the medical and surgical bariatric multidisciplinary team.

Bariatric physician in primary (can be the general practitioner) or secondary care (usually a diabetologist)
Dietician
Specialist nurse
Appropriately trained mental health professional
Bariatric surgeon
Anaesthetist
Radiologist
± Exercise therapists
Other secondary care specialties, e.g. respiratory/sleep medicine, cardiology

NICE estimated that about 80% of people fulfilling the eligibility criteria would have no medical or psychological reason why they would not be fit for surgery. An estimate is that perhaps 10% of these might want it, if bariatric/metabolic surgery were to be promoted or recommended by physicians to patients.

PRINCIPLES OF SETTING UP A BARIATRIC/METABOLIC SURGERY SERVICE

As for gastrointestinal cancer surgery where a number of different specialists routinely work together, it is now agreed that 'bariatric physicians/internists', dieticians, mental health professionals and nurse practitioners should be part of the team assessing and managing long-term care after bariatric surgery (***Table 68.4***). The perioperative risk of surgery is low; however, outcomes can be improved further with appropriate multidisciplinary team (MDT) work-up. Using risk scores such as the Edmonton Obesity Staging System (EOSS) and the Obesity Surgery–Mortality Risk Score (OS-MRS) can help the team discuss with patients the likely prognosis without surgery, and the risk of complications. The OS-MRS scores one point for each of: age 45 or more; BMI 50 kg/m^2 or more; male gender (owing to central obesity); hypertension (owing to central obesity); and increased deep vein thrombosis (DVT)/pulmonary embolism (PE) risk. The more points that are present, the greater the risk. Swedish registry data indicate that obstructive sleep apnoea (OSA) is a risk factor for anastomotic leak. Therefore OSA should be actively investigated as treatment with continuous positive airway pressure might reduce risk. Poorly controlled diabetes must also be considered a risk factor, as it is for all other operations.

Better surgical results are likely in high-volume surgical units. IFSO and the American Society for Metabolic and Bariatric Surgery (ASMBS) recommended 100–125 cases per year, and there should be at least two surgeons each performing 50 or more cases. Surgeons early in the learning curve (about 100 for gastric bypass) need to be mentored before independent practice. Irrespective of the technical expertise of each surgeon, higher volumes usually mean that there are sufficient

MDT members available to provide support and follow-up. Each unit should have expertise in a variety of surgical procedures, including revisions.

It is routine to put patients on a 'liver shrinkage diet' for at least 2 weeks before surgery, especially when there is central obesity, as this is associated with a large liver that can make surgery impossible. Male patients, especially those with central obesity, a very dense/hard abdomen, OSA/diabetes and BMI >50 kg/m^2, may need more, supervised, mandatory weight loss to make surgery safe.

Although not stipulated in any national guidance, every successful bariatric unit depends on active patient support groups and preoperative education sessions, which are best run by bariatric nurses and dieticians. These are invaluable in preparing patients for surgery, and it is difficult to conceive how a programme can run without them. Also, the ward and outpatient environment must be suitably equipped for patients with severe obesity.

Summary box 68.3

Multidisciplinary assessment

- Every patient should be assessed and managed by a coherent and well-functioning team of healthcare professionals with varied backgrounds and expertise
- Improved outcomes are usually achieved in high-volume specialised units
- Data collection and submission to national registries are recommended to provide quality assurance and long-term outcomes

Laparoscopic surgery and enhanced recovery

Bariatric surgery has been transformed by its amenability to laparoscopic techniques, including intracorporeal suturing and modern laparoscopic stapling devices. Probably equally important is the adoption of enhanced recovery after surgery (ERAS) protocols, with preoperative education, avoidance of catheters, central venous and arterial lines and early mobilisation. Free access to fluids is routine immediately after surgery and, because of the relative lack of pain, patients can mobilise straightaway. It is usual for gastric bypass and sleeve gastrectomy patients to go home on postoperative day 2 or 3. Gastric banding patients can be treated as day cases or can go home within 24 hours. The main cause of death after surgery is DVT/PE rather than anastomotic leakage or bleeding; appropriate prophylaxis is usually used for at least a week.

Randomised controlled trial evidence for the different types of surgery

Some evidence suggests that weight loss is more with a Roux-en-Y gastric bypass than sleeve gastrectomy. However, bariatric surgery needs more RCTs comparing different procedures with long-term follow-up. One of the challenges is keeping the research question relevant given the rapidity with which surgeons adopt different techniques. Long-term, large-scale, pragmatic RCTs with good follow-up are needed to inform practice. The By-Band-Sleeve study is an ongoing pragmatic, multicentre, three-arm RCT in the UK assessing weight loss and quality of life at 3 years between gastric bypass, gastric banding and sleeve gastrectomy in 1341 patients; it is expected to report in 2023.

Summary box 68.4

Evidence for the different operations

- Usually the operation choice is guided by patient, surgeon and unit preference
- Well-conducted RCTs comparing the different operations, with comprehensive follow-up, are needed
- Bariatric surgeons and bariatric surgical units should be recording their outcomes, ideally in a registry

The common operations

According to the IFSO Global Registry, in 2018 sleeve gastrectomy constituted 46%, gastric bypass 38%, one-anastomosis gastric bypass procedures 7.6% and gastric banding 5% of procedures. Other procedures include the biliopancreatic diversion (BPD) procedure and its duodenal switch variant (BPD/DS). The variety of procedures usually reflects surgeons' expertise and surgeons' and patients' preferences, as there are no RCTs beyond 5 years comparing different operations. Sleeve gastrectomy is now the most common operation and has gained rapid popularity at the expense of gastric banding and to a lesser extent gastric bypass. Some clinical outcomes are shown (*Table 68.5*). The mechanism of action of most weight-loss procedures remains incompletely understood.

TABLE 68.5 Malabsorption, per cent excess weight loss (% EWL)[a] and diabetes remission after bariatric surgery.

	Protein/ calorie malabsorption	3-year % EWL	3-year % diabetes remission
Sleeve gastrectomy	No	50–60%	50%
Gastric bypass	No	50–60%	50%
OAGB	Yes	60–80%	80%
Gastric band	No	40–50%	20%
BPD/DS, SADI-S	Yes	70–80%	80%

BPD, biliopancreatic diversion; DS, duodenal switch; OAGB, one-anastomosis gastric bypass; SADI-S, single-anastomosis duodenoileal bypass with sleeve gastrectomy.

[a] %EWL refers to the excess weight lost above a notional upper normal body mass index of 25 kg/m^2. Per cent weight loss is another way of measuring weight change, preferred by physicians.

César Roux, 1857–1934, surgeon, Lausanne, Switzerland.

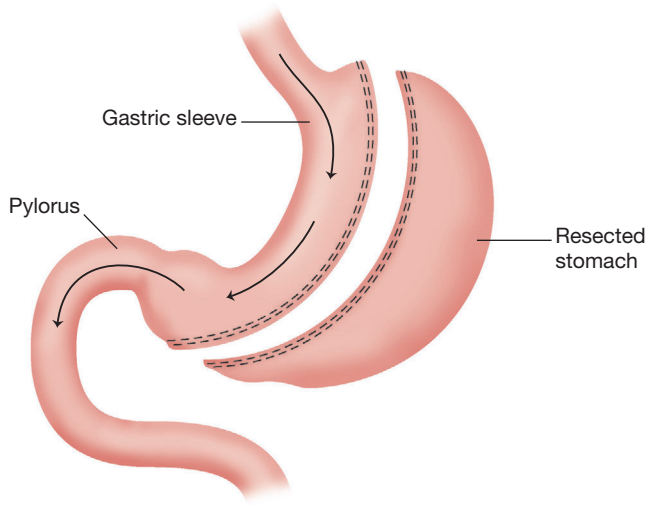

Figure 68.1 Sleeve gastrectomy. (Adapted from Griffin SM, Raimes SA, Shenfine J. *Oesophagogastric surgery*, 5th edn. London: Saunders Elsevier, 2013.)

Reduced appetite and early satiety are common features that are potentially explained by changes in levels of gut hormones such as peptide YY (PYY) and GLP-1 and how these interact with the brain.

Sleeve gastrectomy

Sleeve gastrectomy (***Figure 68.1***) is less challenging to perform than gastric bypass. It evolved from the magenstrasse and mill operation, in which the divided fundus (the 'mill') was left in continuity with the lesser curve-based tube (the 'main street'). Initially, it was done as the first step of a duodenal switch operation; however, it was found to be effective on its own without the switch (see ***Biliopancreatic diversion/duodenal switch***).

The lesser curve-based gastric tube is constructed over a size 32–36Fr bougie, although some surgeons advocate use of larger sizes to reduce the risk of staple line leakage. Linear stapling devices are used. There is variation in the techniques employed between how wide the staplers should be and whether reinforcement strips should be used. The Achilles heel of the sleeve is the risk of a staple line leak at the angle of His, which can take months to heal owing to the high-pressure system in the stomach with an intact pylorus. Another concern in the long term is symptomatic reflux and *de novo* Barrett's oesophagus (see ***Chapter 66***). A proportion of patients will need revisional surgery in future for weight regain.

The mechanism of action is still being investigated. The initial belief that sleeve gastrectomy acts as a restrictive procedure has been challenged by studies which that show gastric emptying is accelerated rather than delayed after sleeve gastrectomy. A change in satiety gut hormones and bile salt metabolism, similar to those described after gastric bypass, may explain some of the phenomena observed.

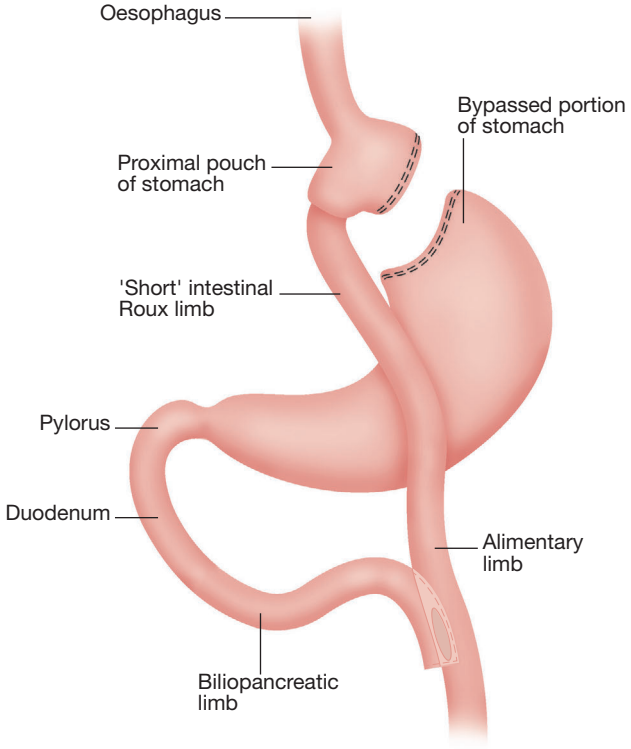

Figure 68.2 Gastric bypass showing a short vertical lesser curve-based gastric pouch with a Roux-en-Y jejunojejunostomy reconstruction. (Adapted from Griffin SM, Raimes SA, Shenfine J. *Oesophagogastric surgery*, 5th edn. London: Saunders Elsevier, 2013.)

Roux-en-Y gastric bypass

Despite the variety in laparoscopic techniques described and the lack of standardisation, most agree that Roux-en-Y gastric bypass (***Figure 68.2***) should include a short vertical lesser curvature-based gastric pouch. The techniques available for construction of the pouch-jejunostomy are linear stapler with suture closure of the defect, circular stapler and entirely hand sewn. It is routine to perform a leak test. The Roux limb can be retro- or antecolic. There is no standard length of the biliary and Roux limbs; however, the biliary limb is usually kept short to reduce vitamin and mineral deficiencies and the Roux limb length is varied between 100 and 150 cm. There are no consistent data regarding the effect of different limb lengths on weight loss. Bowel continuity is restored by a 'Y' jejunojejunostomy, which is either stapled with suture closure of the defect or stapled in its entirety.

It is now recognised that the mechanism of action is complex. Patients lose weight, at least in part, because they eat less owing to a change in appetite, which is facilitated by a change in satiety gut hormones. Other mechanisms such as changes in energy expenditure and change in food preferences may also play a role.

Wilhelm His, 1831–1904, Professor of Anatomy, Leipzig, Germany.
Norman Rupert Barrett, 1903–1979, surgeon, St Thomas' Hospital, London, UK.

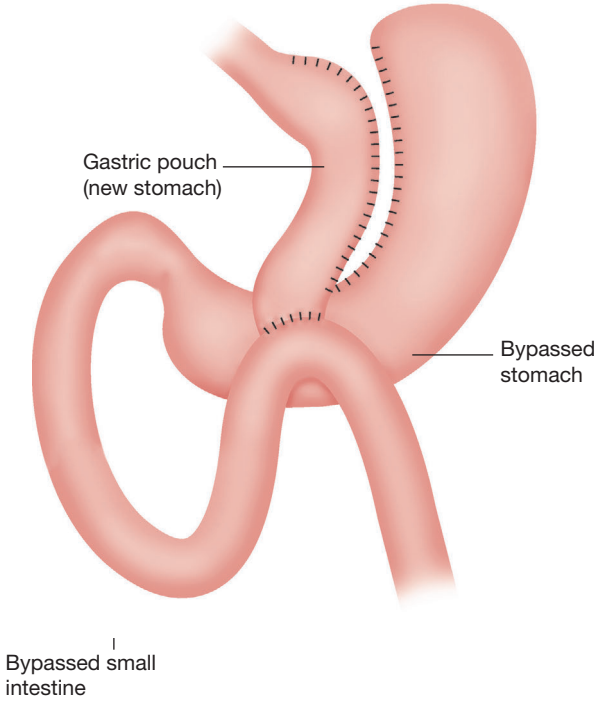

Figure 68.3 Gastric bypass showing a longer vertical lesser curve-based gastric pouch with gastrojejunostomy reconstruction (one-anastomosis gastric bypass).

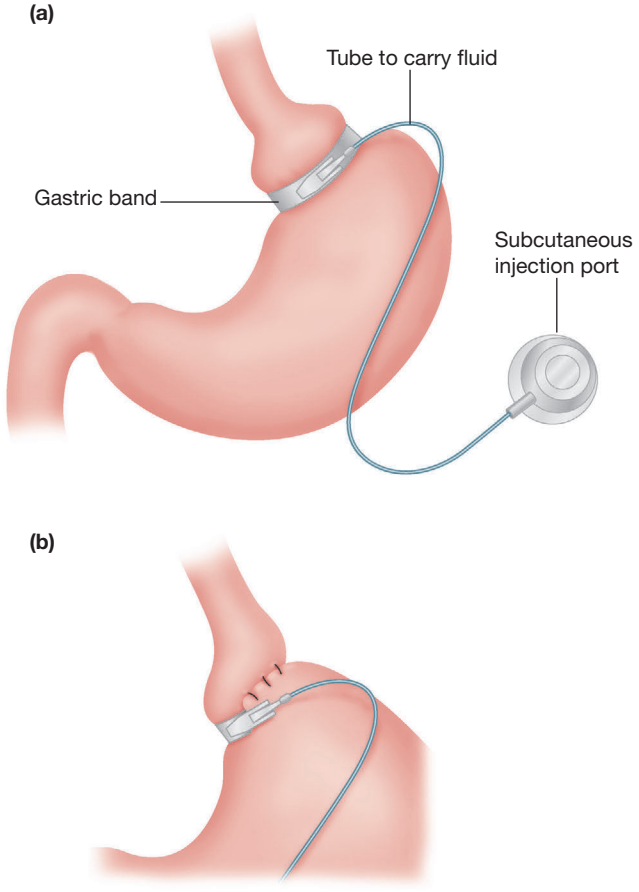

Figure 68.4 Adjustable gastric band. Gastric band surgery showing **(a)** a small 'virtual' pouch of stomach below the gastro-oesophageal junction and **(b)** gastrogastric tunnelling sutures. (Adapted from Griffin SM, Raimes SA, Shenfine J. *Oesophagogastric surgery*, 5th edn. London: Saunders Elsevier, 2013.)

One-anastomosis gastric bypass

One-anastomosis gastric bypass (OAGB) (***Figure 68.3***), previously known as a mini-gastric bypass, was first described by Rutledge. The objective was to develop a technique that is technically less demanding with only one anastomosis (antecolic loop gastrojejunostomy without a Roux-en-Y configuration) and a longer gastric pouch than for standard gastric bypass. Similar weight loss outcomes have been reported but there is concern regarding symptomatic biliary reflux causing gastritis or oesophagitis, marginal ulcers and the management of anastomotic leaks owing to a potentially high volume of biliary and pancreatic secretions. With the Roux-en-Y historically being the standard in surgery of the stomach for ulcer disease and cancer, there is further concern owing to possibly increased risk of Barrett's oesophagus and gastric or oesophageal cancer associated with biliary reflux. These outcomes will need long-term investigation.

Gastric banding

Although use of adjustable gastric banding (***Figure 68.4***) is declining, it did boost the popularity of bariatric surgery because of perioperative safety, lack of nutritional complications and relative ease and availability. The pars flaccida technique (through the window of the lesser omentum) is now standard practice with a band placed just below the oesophagogastric junction, making a small 'virtual' gastric pouch. The band is sutured into place anteriorly with gastrogastric tunnelling sutures to reduce slippage. The access port is routinely sutured to the rectus sheath in the upper abdomen for ease of access by a non-coring, Huber needle for band adjustments.

The operation appears to work by reducing hunger, probably vagally mediated. The initial surgical placement is only the beginning of the treatment. Specialist nurses, physicians and surgeons do 'band consultations' to assess eating habits and then perform an adjustment with injection or aspiration of saline if indicated. The objective is to reach the so-called 'sweet spot' of optimal appetite control. Follow-up should be monthly to begin with as needed during the first year, with full MDT support to help patients get the best use out of their bands. Lack of appropriate follow-up is why results in the literature vary so much, with a consequent high band removal rate.

Biliopancreatic diversion/duodenal switch

BPD, described by Scopinaro, produces greater weight loss than other procedures but is associated with a higher

Robert Rutledge, contemporary, surgeon, Las Vegas, NV, USA.
Nicola Scopinaro, 1945–2020, Professor of Surgery, Genoa, Italy.

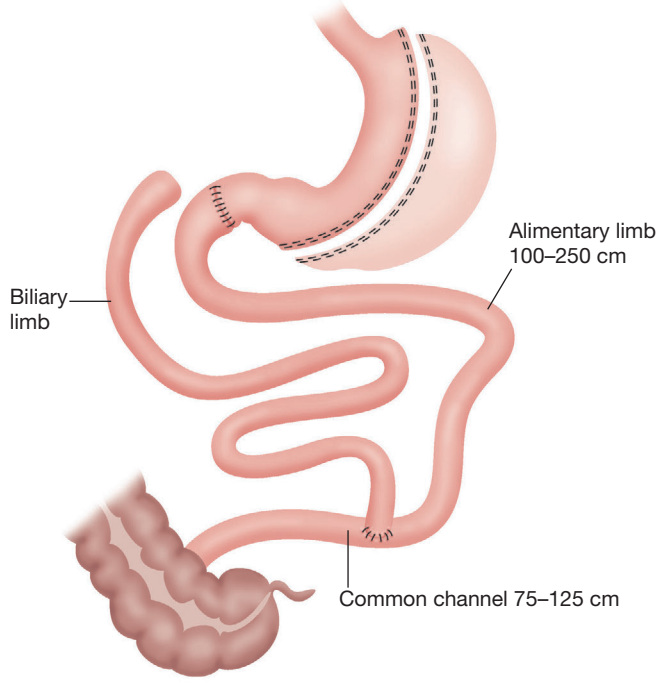

Figure 68.5 Biliopancreatic diversion with duodenal switch variant (BPD/DS).

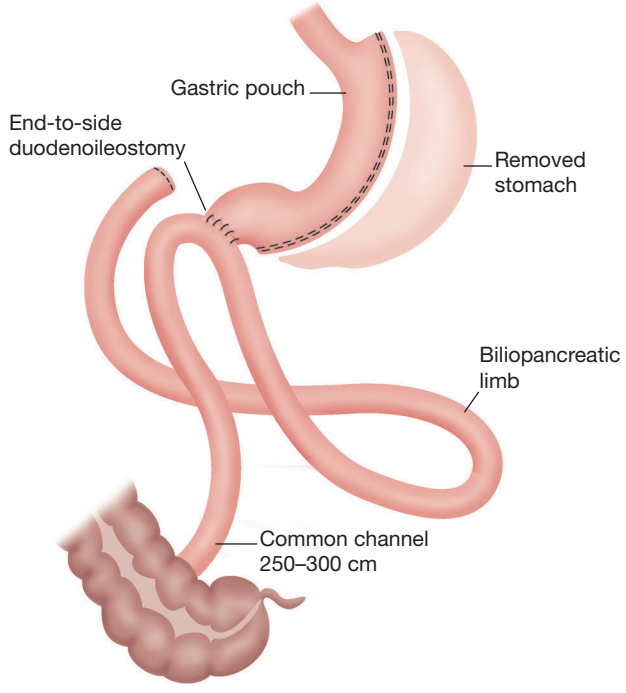

Figure 68.6 Single-anastomosis duodenoileal bypass with sleeve gastrectomy (SADI-S).

nutritional complication rate. The mechanism of action appears to be mainly malabsorption of calories. BPD/DS is the version mainly performed (***Figure 68.5***). A sleeve gastrectomy is followed by division of the duodenum just distally to the pylorus. The ileum is divided with a linear stapler, followed by a duodenoileostomy and ileoileostomy with the objective of creating a common channel of 75–125 cm and an alimentary channel of 100–250 cm. The long remaining biliary limb is not measured.

BPD/DS is increasingly seen as a definitive procedure, particularly after significant weight regain following sleeve gastrectomy. A high-protein diet and regular vitamin and mineral supplements with lifelong monitoring and patient commitment, to avoid malnutrition, is essential postoperatively. Only a few centres offer these procedures.

Single-anastomosis duodenoileal bypass with sleeve gastrectomy

Single-anastomosis duodenoileal bypass with sleeve gastrectomy (SADI-S) is a novel procedure based on the BPD/DS. A sleeve gastrectomy is followed by an end-to-side duodenoileal anastomosis (***Figure 68.6***). The length of the common channel–alimentary limb is 250–300 cm. Potential advantages include the preservation of the pylorus, elimination of one anastomosis compared with the duodenal switch and reducing operating time and risk of perioperative complications.

Complications

The common complications are shown in ***Table 68.6***. In sleeve gastrectomy, a staple line leak at the angle of His usually presents any time *after* discharge up to 30 days, and patients can also deteriorate rapidly with sepsis. Urgent computed tomography (CT) scanning and relaparoscopy is indicated, with source control by drainage the major goal. Patients are typically in hospital for months and need multiple reinterventions, including any of: endoscopic interventions (stenting, endoscopic vacuum therapy), making a controlled fistula, conversion to gastric bypass and fistula enterostomy. Long-term nutritional support is needed as patients are severely catabolic after complications from both bypass and sleeve surgery.

Anastomotic leakage, bleeding and closed loop obstruction after Roux-en-Y or one-anastomosis gastric bypass can be life-threatening. If a bypass patient is not well after 24 hours urgent consideration should be given to oral contrast X-ray swallow or CT scanning and/or relaparoscopy. Other than a feeling of 'impending doom' patients may have few overt features of sepsis and abdominal examination can be very misleading. Deterioration after an anastomotic leak can be very rapid and there is no time for delay.

Very few patients with gastric bands develop early intra-abdominal complications. Unfortunately a large number of patients have their bands removed later on if there is inadequate follow-up, a late complication or the patient is unable to tolerate the device.

The incidence of late complications is difficult to estimate as so many patients are lost to follow-up. Internal hernias develop as weight is lost and hernia spaces open up after gastric bypass. CT scanning has a high rate of false negatives for internal hernia, so anyone presenting with severe, cramping abdominal pain 2–3 years after surgery needs to be

TABLE 68.6 Estimated early surgical complication rates, operative mortality after sleeve gastrectomy, Roux-en-Y or one-anastomosis gastric bypass, and gastric banding, and late complications.

	Early	Mortality	Late
Sleeve gastrectomy	Leak at angle of His (1–2%)	0.1%	Gastro-oesophageal reflux
	Intra-abdominal bleed (2–3%)		Barrett's oesophagus
	DVT/PE (<1%)		Weight regain
Gastric bypass	Anastomotic leak (<1%)	0.1%	Internal hernia
	Intra-abdominal bleed (2–3%)		Chronic abdominal pain
	Unspecified obstruction (1–2%)		Malnutrition if long limb bypass
	DVT/PE (<1%)		Anastomotic ulcer/stricture
			Weight regain
Gastric band	Access port infection (1%)	0.05–0.1%	Band infection
	DVT/PE (<0.1%)		Tubing leak
			Slippage
			Erosion into stomach
			Band intolerance
			Failure to lose weight/weight regain

high priority for investigation by laparoscopy. Closure of the internal hernia spaces is now standard of care in Roux-en-Y gastric bypass.

Summary box 68.5

Acute complications

- Anastomotic leak and staple line dehiscence can be rapidly fatal and require emergency laparoscopy
- Internal hernias developing after surgery are very difficult to diagnose other than by prompt laparoscopy; they require a high index of suspicion

Outcomes reported

There is wide variation in how surgeons report the results of surgery, which means that it is often difficult to compare studies. There is a need to standardise clinician-reported outcomes and patient-reported outcome measures (PROMs) into an agreed core outcome set that includes risk stratification. Obvious PROMs include quality of life.

Follow-up and a shared care model of chronic disease

Shared care arrangements with surgeons/physicians and primary care need to be in place so that diabetes and hypertension medications and dosage can be appropriately reduced as weight is lost. Every patient with diabetes needs at least an annual review.

Although sleeve gastrectomy, short-limb forms of gastric bypass and gastric banding do not cause protein-calorie malabsorption, bariatric surgery can cause severe vitamin and mineral deficiencies, amplifying pre-existing deficiencies caused by obesity. All patients should have lifelong routine metabolic and nutritional monitoring (*Table 68.7*). Patients need regular multivitamins/trace element supplements (*Table 68.8*). The minimum frequency of assessment is 3–6 monthly in the first postoperative year, 6–12 monthly in the second year and at least annually thereafter. Folic acid supplementation should be considered in all sexually active women of childbearing age because of the risk of neural tube defects. This is especially important as fertility often improves after surgery. The MDT also needs to support the small number of patients who develop severe mental health issues after surgery as there is a slightly increased risk of suicide after gastric bypass.

TABLE 68.7 Summary of British Obesity and Metabolic Surgery Society (BOMSS) biochemical guidance after bariatric surgery.

Blood tests all patients should have at baseline
Full blood count, including haemoglobin, ferritin, folate and vitamin B12 levels, urea and electrolytes, liver function tests, vitamin D, Ca^{2+}, parathormone, HbA1c, lipid profile
Postoperatively
After gastric banding: Annual full blood count, urea and electrolytes, HbA1c, fasting glucose, lipids as appropriate *After sleeve gastrectomy, forms of gastric bypass, BPD/DS, SADI-S:* As for banding + liver function tests, ferritin, folate, vitamin D, Ca^{2+}, parathormone at 3, 6, 12 months then annually; vitamin B12 at 6, 12 months then annually; zinc, copper annually; vitamins A, E, K, selenium if concern (e.g. steatorrhoea, night blindness, unexplained fatigue, anaemia, metabolic bone disease, chronic diarrhoea, heart failure)

BPD, biliopancreatic diversion; DS, duodenal switch; HbA1c, glycated haemoglobin; SADI-S, single-anastomosis duodenoileal bypass with sleeve gastrectomy.

TABLE 68.8 Summary of British Obesity and Metabolic Surgery Society (BOMSS) nutritional and micronutrient guidance after bariatric surgery.

After gastric banding
Multivitamin and mineral supplement, thiamine if vomiting, vitamin D, iron
After sleeve gastrectomy, forms of gastric bypass, BPD/DS, SADI-S
As for banding + selenium, copper, zinc, folic acid, vitamins B12, A, E, K BPD/DS, SADI-S may require higher doses

BPD, biliopancreatic diversion; DS, duodenal switch; SADI-S, single-anastomosis duodenoileal bypass with sleeve gastrectomy.

Summary box 68.6

Shared care model of chronic disease

- Close collaboration between surgeons, physicians and primary care doctors is needed to enable seamless follow-up before and after surgery with a focus on the long-term care of patients
- Patients should be committed to lifelong vitamin and micronutrient monitoring and replacement

FUTURE CHALLENGES

Patients with obesity suffer from widespread prejudice. Understanding that the obesity epidemic currently experienced in different parts of the world is driven by a change in the environment towards becoming 'obesogenic' and not a lack of willpower would be the first step in removing the barriers to more surgery.

All surgeons should contribute their results to national registries so that safety can be monitored and operation trends established. Ideally registries should also link to other national healthcare records, e.g. diabetes databases, so that long-term outcomes data can be collected outside of funded RCTs.

FURTHER READING

Adams TD, Gress RE, Smith SC *et al.* Long-term mortality after gastric bypass surgery. *N Engl J Med* 2007; **357**: 753–61.

Mingrone G, Panunzi S, De Gaetano A *et al.* Bariatric–metabolic surgery versus conventional medical treatment in obese patients with type 2 diabetes: 5 year follow-up of an open-label, single-centre, randomised controlled trial. *Lancet* 2015; **386**: 964–73.

National Institute for Health and Care Excellence. *Obesity: identification, assessment and management.* NICE Clinical Guideline 189. London: NICE, 2014. Available from https://www.nice.org.uk/guidance/cg189.

O'Kane M, Parretti HM, Pinkney J *et al.* British Obesity and Metabolic Surgery Society Guidelines on perioperative and postoperative biochemical monitoring and micronutrient replacement for patients undergoing bariatric surgery: 2020 update. *Obes Rev* 2020; **21**: e13087.

Sjöström L. Review of the key results from the Swedish Obese Subjects (SOS) trial – a prospective controlled intervention study of bariatric surgery. *J Int Med* 2013; **273**: 219–34.

Welbourn R, Dixon J, Barth JH *et al.* NICE-accredited commissioning guidance for weight assessment and management clinics: a model for a specialist multidisciplinary team approach for people with severe obesity. *Obes Surg* 2016; **26**: 649–59.

Welbourn R, le Roux CW, Owen-Smith A *et al.* Why the NHS should be doing more bariatric surgery; how much should we do? *BMJ* 2016; **353**: i1472.

Welbourn R, Pournaras DJ, Dixon J *et al.* Bariatric surgery worldwide: baseline demographic description and one-year outcomes from the second IFSO Global Registry Report 2013–2015. *Obes Surg* 2018; **28**: 313–22.